

Question no. 1 Answer

Virtualization is the process of creating a virtual version of something - like a server, storage device, network or operating systems - so multiple systems can run on a single physical machine.

Type 1 Hypervisor (Bare-metal)

- Installed directly on physical hardware.
- Doesn't need a host OS.
- More efficient and secure
- Example: VMware ESXi, Microsoft Hyper-V, Xen.

Type 2 Hypervisor (Nested)

- Runs on top of a host operating system.
- Easier to install but slightly less efficient.
- Suitable for personal or testing use.
- Example: Oracle Virtualbox, VMware Workstation.

Question no. 2 answer

Virtual clustering refers to the use of virtualization technologies to create a logical cluster of virtual machines (VMs) across multiple physical servers. These virtual machines can work together as a unified resource pool to handle tasks or applications, even though they are running on separate physical machines.

Key Elements of Virtual Clustering:

i) Virtual Machines (VMs):

Each VM acts as a node within the cluster. VMs can run different operating systems and applications while sharing resources from

physical host.

(ii) Virtual Switches: These are used to create a network layer between VMS, allowing communications across VMS as if they are on same network.

(iii) Resource Pooling: Resources such as CPU, memory, and storage are pooled from multiple physical machines and allocated to VMS based on demand.

Question no. 3, Answer

Feature	Grid Computing	Cloud Computing
Purpose	High performance computing for specific tasks.	On-demand access to a broad range of IT services.
Resource Pooling	Distributed resources from multiple organizations.	Centralized resources managed by a provider.
Scalability	Limited, depends on available grid resources.	Highly scalable with automated management.

owner-ship	Distributed across different entities	- Managed and owned by service providers.
flexibility	Less flexible	- Highly flexible.
example	SETI@home, Scientific Simulations	AWS, Google Cloud, Microsoft Azure.

Question NO. 4 Answer

i) Network virtualization

It is the process of combining multiple physical networks into a single virtual network. It abstracts and isolates the network resources to provide a flexible, manageable, and scalable network environment.

Advantages:

➤ Resource Efficiency:

Better utilization of network resources by running multiple virtual networks over the same physical infrastructure.

➤ Scalability:

easily expand and modify network

configurations without impacting physical hardware.

Flexibility:

Dynamic configuration of virtual networks to meet specific application or business needs.

Example:

- VMware NSX and Cisco ACI offers solutions for creating and managing virtual networks on top of physical network infrastructure.

ii) Storage Virtualization

Storage Virtualization is the process of abstracting and pooling multiple storage devices into a single virtualized storage resource.

Benefits:

→ Simplified Management:

Storage resources are managed as a single entity, making it easier to scale and allocate space.

→ Improved Utilization:

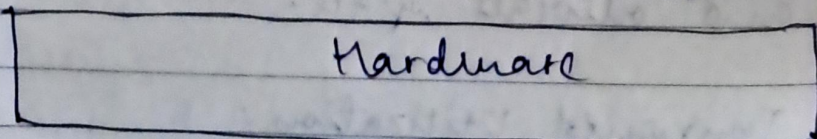
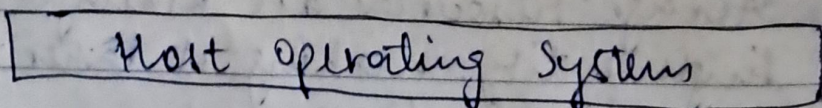
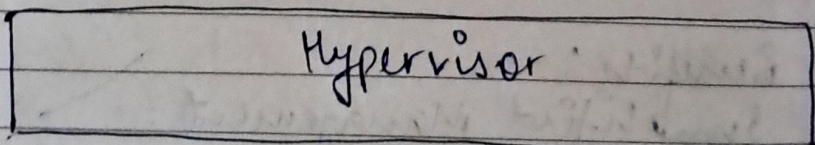
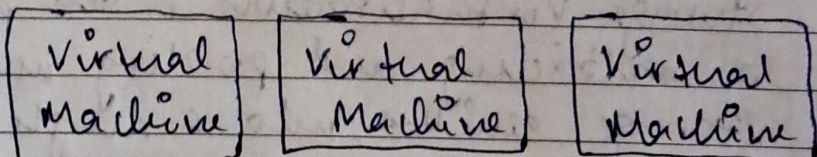
Optimizes storage by dynamically allocating space based on usage patterns.

- **Redundancy and Failover:**
Virtual storage can offer features like data replication and automatic failover, enhancing data availability.

Example:

- VMware vSAN and Microsoft Storage Spaces are examples of solutions that provide storage virtualization for better management and scalability.

Question No. 5 Answer



Virtualization Architecture

Virtualization Architecture has the following components:

- i) **Hardware Layer**
At the very bottom, you have physical hardware (CPU, memory, storage).
- ii) **Host Operating System (OS):**
This is the standard OS running directly on the hardware.
- iii) **Hypervisor (Virtual Machine Monitor):**
This layer sits directly above the host OS. It is responsible for managing VMS and allocating physical resources to each VM.
- iv) **Virtual Machines (VMS):**
These are isolated environments, each running its own guest OS and applications.